Group 4

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Project Proposal

True and Position Guess Game

Think, guess, and outsmart—two players, one challenge.

1. Problem Statement

In today's mobile gaming environment, most applications focus primarily on entertainment, offering fast-paced visuals and rewards but providing little opportunity for mental development. Many existing number-guessing or logic-based games are single-player and lack the interactive challenge that encourages players to think strategically or compete intellectually with others. As a result, users who enjoy puzzle-solving and logical reasoning have limited options for engaging, educational two-player games that are both simple and stimulating. This gap reduces opportunities for users to practice analytical thinking and friendly competition in a mobile environment designed for learning and fun.

Our target users include students, puzzle enthusiasts, and casual gamers who enjoy interactive, logic-based challenges. They seek mentally engaging mobile games that are easy to use, competitive, and educational providing both entertainment and opportunities to strengthen their reasoning and focus in a social two-player environment.

2. Proposed Solution

The proposed solution is the development of an Android-based two-player logic game called True & Position Guess Game. This mobile application is designed to combine entertainment with mental stimulation by engaging players in a competitive number-guessing challenge. Each player secretly selects a number, and they take turns guessing each other's number. After every guess, the system provides instant feedback: "True" indicates correct digits regardless of position, while "Position" shows digits correctly placed. The game continues until one player successfully guesses all digits in the correct positions.

Through this structured feedback system, the app encourages players to apply logical reasoning, pattern recognition, and strategic thinking to narrow down possibilities with each turn. The user-

friendly interface and real-time interaction make it accessible for students, puzzle lovers, and casual gamers who want to test and improve their analytical skills while competing with friends.

3. Core Features (Minimum Viable Product – MVP)

- Two-Player Turn-Based Gameplay – Allows two users to take turns guessing each other's secret numbers on the same device, with automatic turn switching in given time.
- Instant Feedback System – Displays "True" (correct digits) and "Position" (correctly placed digits) after each guess to guide player strategy.
- Input Validation – Ensures valid numeric entries with no repeated digits, no invalid lengths, and proper formatting.
- Winning Condition & Game Result Display – Detects when a player has correctly guessed all digits and shows both players' secret numbers with a game summary.
- Simple and User-Friendly Interface – Provides a clean, intuitive layout for easy interaction, suitable for both beginners and experienced users.

How it works

- Open the App

  The app loads the home screen with a clean interface showing to start the game.
- Player Setup

  The app prompts the first player to choose a secret number (3–6 digits, depending on difficulty).

  Input validation ensures the number has no repeated digits and meets formatting rules.
- Hiding the Number

  Once entered, the secret number is stored securely in the app and hidden from the second player.

  The screen then prompts the second player to enter their secret number without revealing the first player's number.
- Turn Initialization

  After both secret numbers are set, the app indicates whose turn is first (e.g., Player 1). The game interface displays input fields for guesses, a log area for previous guesses, and feedback counters.

- Game Ready to Play

  Player 1 makes their first guess, and the app provides instant “True” and “Position” feedback.

  The turn then automatically switches to Player 2, continuing the gameplay cycle.

- Finally, after the guesses the winner is announced